

Abby Williams, PhD

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PROFILE

Senior data scientist and builder who works on workforce and labor market intelligence in production. Designed and deployed Meridian, a platform that classifies career trajectories from 70,000+ human capital records, augments them with a real-time labor market signal via the Adzuna API, and serves 5,000+ users, built with NLP, ensemble modeling, and a validated GPT-4 layer. Leads a 10-person ML and product team, and brings PhD-level depth in NLP, time series, and large-scale behavioral analysis, earned while building and shipping these systems in the same years.

TECHNICAL SKILLS

Languages: Python, R, SQL, C/C++, Regex, Bash

ML / Modeling: scikit-learn, PyTorch, TensorFlow, XGBoost, Random Forest, K-means, KNN, logistic regression, PCA, EFA, time-series forecasting, interrupted time series, Bayesian methods, SHAP explainability

NLP: TextBlob, VADER, sentiment scoring, open-text feature engineering

Data Engineering: ETL pipelines, data modeling, schema design, validation, normalization, batch processing, feature engineering

Orchestration/DevOps: Streamlit, Airflow, Docker, REST APIs, CI/CD

Cloud & Databases: AWS (S3, EC2, ECS, Lambda, ECR, ALB, CloudWatch), GCP (BigQuery, AI Platform), Microsoft Fabric, PostgreSQL, MySQL, MongoDB, data lakes

Analysis/Visualization: pandas, NumPy, Tableau, Power BI, Matplotlib, Jupyter, Qualtrics, SPSS, STATA, ArcGIS

People Analytics: behavioral segmentation, predictive modeling, talent intelligence, skills taxonomy, longitudinal and survey analysis, psychometrics, NPS and engagement modeling

WORK EXPERIENCE

New York University

New York, NY

Senior Data Scientist & Product Lead (Assistant Dean), NYU GSAS

Sep 2022 - Present

- Built and now lead a 10-person team across applied ML, data engineering, and product. Set the technical standards the team works to, review model outputs before they ship, and partner with senior leadership on experiment design and the metrics models are held to.
- Designed and deployed *Meridian*, owning feature engineering, model selection, validation, and AWS deployment for a career intelligence platform.
- Deployed a containerized application on AWS ECS Fargate with autoscaling for reliable performance under load.
- Built CI/CD pipelines with Git-based workflows that cut release cycles from 4 hours to ~8 minutes while improving deployment reliability, monitoring, and rollback.
- Built a multi-source ETL pipeline in Python and SQL to ingest and normalize ~70K records from S3, Excel, and REST APIs, with schema validation and automated error handling that cut data refresh latency from ~45 seconds to under 5.
- Produced four validated behavioral career personas from k-means segmentation using RIASEC composites, satisfaction metrics, and motivational factors.
- Managed 5 large-scale annual survey programs, from instrument design and sampling through automated reporting to senior leadership and external funders.
- Designed an end-to-end enrollment analytics pipeline for graduate program review, from Qualtrics instrument design through Python data cleaning to an interactive Streamlit dashboard. It tracks funnel diagnostics, experience ratings, NPS-equivalent scores, and post-completion compensation outcomes.
- Designed the sarcasm-aware sentiment ensemble behind Meridian's feedback layer, combining VADER and TextBlob with a high-precision, cue-based sarcasm detector to correct the failure mode where standard tools misread candid feedback; the corrected signal feeds a thematic layer by topic and division and contributed to a 12% improvement in user satisfaction.
- Built the professional development office from the ground up with 250+ stakeholders across senior leadership, departments, and student groups, and co-wrote the grant funding the role.

Assistant Director

Mar 2016 - Aug 2022

- Secured \$18M+ in competitive funding across Mellon, NSF, and other major funders; tripled program awards through predictive analytics and targeted outreach, tied to a \$3M grant renewal.
- Built automated data tracking infrastructure that cut fellowship processing time 50% and improved reporting accuracy across concurrent award programs.
- Designed the office's first real-time Tableau dashboard, and used StoryMaps and ArcGIS to map where applicants came from and where awards concentrated geographically.
- Built a CRM to manage Fulbright leads and pre- and post-award processes, centralizing communications and improving placement tracking across programs.
- Built the Public Humanities Certificate from zero: \$2.5M initial grant, curriculum design, 4x enrollment growth, top 10 nationally in Fulbright winners over a decade.

- Led the Luce Foundation scholarship review as team lead, managing four reviewers and delivering recommendations to Luce senior staff.

Program Administrator

Jan 2015 - Feb 2016

- Ran user research with 150+ PhD users using ATLAS.ti, applied findings to redesign service delivery, reducing in-person visits 60%.
 - Streamlined dissertation submission for 300+ doctoral candidates annually; built budgeting systems improving financial oversight accuracy.
- Built a custom fellowship application portal in SurveyMonkey Apply, migrating the office from Google Forms to a fully automated end-to-end process, reducing manual handling and creating a structured, trackable applicant pipeline for the first time.

Handshake AI / Feather (OpenAI)

AI Training Specialist |

November 2025 - April 2026

- Provided expert evaluation and feedback on AI model outputs for career-development use cases, with domain-grounded labeling and rubric design for higher education and PhD pathways.

SELECTED PROJECTS

Meridian Career Intelligence Platform | New York University | Python, NLP, GPT-4, AWS | *January 2025 - Present*

- Built an **ensemble classification and NLP system with GPT-4 integration**: a scikit-learn TF-IDF pipeline feeding XGBoost, Random Forest, logistic regression, SVC, and Naive Bayes with dynamic weighting; sentiment analysis with a custom sarcasm-aware lexicon; a real-time labor market signal via the Adzuna API; and GPT-4 advisor-facing output validated against rule-based guardrails with LLM drift monitoring. *Streamlit dashboard deployed on AWS EC2; live at gsasmeridian.com. Manuscript targeting Journal of Higher Education. <https://gsasmeridian.com/>*

CV-to-Resume Conversion Tool | New York University | 2025 - 2026

AI-driven system that converts 5–10-page academic CVs into structured industry resumes and extracts normalized skill and experience signals, reducing resume preparation time ~70% and accelerating job application readiness for PhD candidates. Streamlit/AWS.

Sentiment-Augmented Persona Validation | New York University | 2024

Cross-validated four behavioral career personas against 60+ interview transcripts and large-scale open-text sentiment, triangulating qualitative and quantitative signal to confirm cluster assignments. Built automated stakeholder dashboards linking sentiment trends to personas and program interventions.

COVID-19 PhD Career Impact Study | NYU / Harvard Dataverse | 2020 - 2022

Interrupted time series analysis on national survey data (n=329) measuring pandemic-driven shifts in doctoral career aspirations. Methods: composite index creation, statistical weighting, longitudinal panel design. Published Journal of Experimental Political Science (2022). Full replication package on Harvard Dataverse. doi:10.1017/XPS.2020.34

NSF/Mellon Career Pathways Dashboard | New York University | 2023 - 2024

Integrated ML persona predictions with demographic and geospatial data across Tableau, ArcGIS, and GCP BigQuery. Visualizations directly informed \$1.2M+ in program funding allocation decisions.

Professional Development Chat Support | New York University | 2024

Built a chatbot to handle routine professional-development questions from students and staff, cutting repetitive email volume.

Public Humanities Program Impact Study | New York University | 2022 - Present

Designed and implemented an RCT to evaluate program effectiveness of a humanities and social science-based graduate internship program for 200+ participants.

EDUCATION

PhD, Administration, Leadership & Technology (Machine Learning & UX focus) | New York University | New York, NY | May 2026

Dissertation: "Designing PhD Career Personas"

The research that became Meridian, developed in parallel with the production system itself. End-to-end ML system built on 70,000+ harmonized cross-sectional survey records. Feature engineering from RIASEC composites, satisfaction metrics, and motivational factors. Model suite: K-means (silhouette=0.37), 3-class XGBoost (F1=0.58, precision=0.60, recall=0.57), Random Forest, logistic regression. SHAP explainability layer for advisor-facing model transparency.

Coursework: Causal Inference, Bayesian Statistics, Deep Learning/Neural networks, Experimental design, Data Visualization, Project Management, Mixed-methods Design, Econometrics, Data Engineering, Gen-AI for Business

MA, International Education | New York University | New York, NY | Dec 2016

BA, Chinese and French | University of Pittsburgh | Pittsburgh, PA | May 2011

LANGUAGES

English (native) | **French** (working proficiency) | **Mandarin** (working proficiency)

SELECTED PUBLICATIONS & SERVICE

- Williams, A. & Williams, F. (2024). *How to Thrive Beyond Academia: Addressing Five Common Challenges for Humanities PhDs*. Public Humanities (Cambridge University Press).
- Cover endorsement. Featured endorsement on Williams, F. (2025) *Options for Success: A PhD's Guide to Navigating Career Transitions and Thriving in Your Next Professional Chapter*. Oxford University Press.
- Haas, N., Gureghian, A., Jusino Díaz, C., & Williams, A. (2022). Through Their Own Eyes: The Implications of COVID-19 for PhD Students. *Journal of Experimental Political Science*, 9(1), 1-21. doi:10.1017/XPS.2020.34
- Speaker, Chronicle of Higher Education - PhD career pathways and academic transitions (with Stacy Hartman)
- Reviewer, Fulbright US Student Program (Taiwan), Boren Awards (Asia), Critical Language Scholarship (Asia), Luce Scholarship (Chair)